



Tiago Barros

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Summary

Infrastructure and network engineer. I build and run on-prem virtualized environments and the distributed systems on top of them, from network and Linux foundations up to the orchestration, automation and observability that keep them running.

Experience

Institute of Telecommunications	Aveiro
Infrastructure & Network Engineer	June, 2024 - Present
• Designed and built two mobile edge datacenters complete with subnetting, pfSense firewalling and routing and VLAN segmentation as well as set up and maintain the virtualized services running on them. • Deploy and manage Proxmox and OpenStack environments and IP networks supporting research and production workloads. • Built the observability and energy-monitoring stack (Prometheus, Grafana, Loki, Telegraf/InfluxDB), including per-outlet power telemetry from PDUs and servers over SNMP/MQTT. • Introduced and maintain the team's internal documentation platform, building its CI/CD delivery pipeline. • Mentor undergraduate students on architecture, code review and operations, including teams contributing to European-funded research projects.	
Research Fellow	May, 2023 - May, 2024
• Researched and deployed multi-edge, multi-slice private 5G networks with the Aether platform. • Developed a GitOps-driven CLI tool for slice deployment and configuration.	

Education

University of Aveiro	Master's
Computer Engineering and Telematics	Sep, 2018 - Dec, 2023
Thesis Title: " Dynamizing Private 5G Edge Deployments "	

Projects

Distributed Network Control Service	September 2025 - Present
Architected a distributed service that automates 5G network operations: owned the system design and the core engineering decisions, which include a message broker for asynchronous actions and gRPC for synchronous queries. Built with OpenTelemetry tracing and observability, and with a stateless design, storing only record logs.	
Internal Documentation Platform (Wiki)	2024 - Present
Initiated and built the team's documentation platform end to end: MkDocs (Material) with a self-hosted GitLab CI/CD pipeline (containerized build, rsync-over-SSH deploy to an nginx LXC). Designed, implemented and maintained.	
Homelab	Oct, 2024 - Present
Built and run a Proxmox-based homelab: OpenWRT-routed dual-stack (IPv4/IPv6) with VLANs, WireGuard VPN for remote access, Nginx as a reverse proxy with TLS (Certbot + Cloudflare DNS) and fail2ban; observability via Prometheus + Loki + Grafana .	

Skills

Infrastructure & Virtualization	Containers & Orchestration
Linux, Proxmox, OpenStack, KVM, LXC, ZFS, Snapshots, Backups	Kubernetes, Helm, Docker, Argo CD, GitOps
Networking	CI/CD & Automation
L2/L3 design, VLANs, Routing, IPv4/IPv6, Firewall, DHCP, DNS, SDN	GitLab CI (self-hosted runners), GitHub Actions, Bash, Systemd
Observability & Telemetry	Programming
OpenTelemetry, Prometheus, Grafana, Loki, Tempo, Telegraf, InfluxDB, SNMP	Go, Python, Bash, C/C++, Java

Languages

English	Portuguese
Highly Proficient	Native